

Contact

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Top Skills

Snowflake Horizon (The new governance standard)

DataOps

Semantic Layer

Languages

English (Full Professional)

Hindi (Full Professional)

Oriya (Native or Bilingual)

Certifications

ITIL V3

Microsoft Certified: Azure Fundamentals

SnowPro Core Certification

Microsoft Certified Technology Specialist: .NET Framework 3.5, Windows Communication Fnd Applications (MCTS)

AWS Certified Solutions Architect

Honors-Awards

Rewarded with Philips Team to Excel award

Awarded with Cognizant Door-Ye Noor

#Awarded with Philips Spotlight

Santosh Tripathy

Staff Engineer | .NET Full Stack | Snowflake | dbt | AWS | FinTech & Healthcare

Charlotte Metro

Summary

Principal Data Engineer | Staff Engineer | Software-First Data EngineeringI don't just build data pipelines; I build AI-ready data foundations with the rigor of high-performance software engineering. With 17+ years across FinTech, Healthcare, and Insurance, I bridge the gap between complex backend architecture (.NET/AWS) and the Modern Data Stack (Snowflake/dbt).Currently, as AVP Tech Lead at LPL Financial, I am focused on the "Software-ization" of Data —applying CI/CD, unit testing, and design patterns to Snowflake Medallion architectures to turn "messy data" into governed, high-velocity business assets. The 2026 Value Proposition:DataOps Mastery: Architected Medallion (Bronze/Silver/Gold) platforms using Snowflake, dbt Core (Semantic Layer), and Airflow, reducing client onboarding from months to days.Cost & Governance (FinOps): Expert in Snowflake Horizon and Protegrity integration. I've delivered a 20% reduction in warehouse operational costs through metadata-driven design and Zero-Copy Cloning.AI-Accelerated Delivery: I don't just use AI; I integrate it. I've boosted team velocity by 40% by rolling out Cursor AI and GitHub Copilot while piloting Agentic AI workflows (LangChain/CrewAI) for automated DataOps and root-cause analysis.Hybrid Architecture: 15+ years of .NET Core & AWS (SAA-C03) means I build the APIs that consume the data. I eliminate BI bottlenecks by serving Gold-layer data directly to custom Angular/React frameworks.# Technical Arsenal:Data: Snowflake (Cortex, Data Share, Snowpipe, Masking), dbt, AWS Glue, Airflow (MWAA).Software: .NET 10/Core, Angular, Microservices, RESTful APIs, SignalR.Cloud/DevOps: AWS (EKS, Lambda, Terraform, EventBridge), GitHub Actions, CI/CD.Next-Gen: LangChain, CrewAI, Vector Databases, Prompt Engineering. Certified: SnowPro Core (COF-C02) | AWS Solutions Architect (SAA-C03) | Azure Data Fundamentals.I am passionate about building "un-layoffable" engineering cultures where AI tools and software best practices meet massive-scale data.Let's connect if you're looking for a Principal or Staff Lead to architect your next-gen data platform.

Experience

LPL Financial

3 years

Assistant Vice President

September 2025 - Present (7 months)

Fort Mill, SC

- Platform Architecture: Architected a Medallion (Bronze → Silver → Gold) data platform on Snowflake, leveraging S3 External Stages and Snowpipe for automated ingestion.
- Semantic Layer & DataOps: Developed complex dbt Core models utilizing incremental materializations and snapshots to build a robust Semantic Layer for financial reporting.
- Data Mesh & Distribution: Implemented Snowflake Data Share to enable zero-copy, governed distribution of curated datasets, eliminating ETL duplication and achieving near-real-time delivery.
- Infrastructure as Code (IaC): Led on-prem to AWS migration using EKS and Terraform to build scalable, cloud-native platform components.
- Advanced Security & Governance: Integrated Protegrity for tokenization and implemented Dynamic Data Masking and RBAC to ensure enterprise-grade PII protection and column-level access control.
- Environment Optimization: Utilized Zero-Copy Cloning to provision instant, isolated environments (Dev/Test/UAT), cutting setup time from days to minutes and reducing storage overhead.
- AI Integration: Leveraged Snowflake Cortex AI (SUMMARIZE, COMPLETE) to embed LLM-powered anomaly explanation and query analysis directly into SQL pipelines.
- Dev Acceleration: Drove a 40%+ improvement in team velocity by championing Cursor AI and GitHub Copilot adoption across the engineering lifecycle.
- Leadership: Mentored global offshore teams and led Agile ceremonies to ensure on-time delivery of high-impact FinTech features

Senior Software Development Engineer

April 2023 - Present (3 years)

Fort Mill, SC

- Led architecture and deployment of distributed applications on AWS EKS, Terraform, and CloudFront, achieving 40% system performance improvement

- Built automated ETL pipelines using dbt, Airflow, and Snowflake, enabling efficient ingestion from on-premise sources and diverse file systems
- Designed ASP.NET Core microservices deployed on Kubernetes for high availability, scalability, and legacy system modernization
- Implemented event-driven integrations via S3 Events, EventBridge, and SNS for real-time data processing, alerting, and downstream automation
- Enforced compliant sensitive data handling through Protegrity tokenization and AWS KMS detokenization workflows
- Delivered Angular UIs on S3 Static Hosting + CloudFront; implemented Infrastructure-as-Code using Terraform for repeatable AWS deployments
- Consolidated CloudWatch alerts into automated Dynatrace + ServiceNow incident creation, reducing MTTR and improving operational monitoring
- Researched and prototyped Generative AI, LLM, and prompt engineering applications for trading and financial advisor platform automation
- Mentored offshore development teams, facilitated sprint planning, and ensured on-time, high-quality delivery of key product features

LSEG (London Stock Exchange Group)

Sr. Consultant

June 2022 - March 2023 (10 months)

Fort Mill, SC

- Led AWS cloud migration translating legacy monolithic workflows and stored procedures into serverless, cloud-native services using Lambda, Athena, S3, EC2, and Glue
- Designed and developed ASP.NET Core RESTful APIs as part of the IBT microservices suite, improving system maintainability and scalability
- Built SignalR hubs with SQL dependency for real-time application notifications, improving operational visibility for business stakeholders
- Delivered Angular UI components for client-facing IBT applications integrated into production workflows

- Integrated all services into Jenkins CI/CD pipelines, enabling streamlined, repeatable deployments
- Collaborated with business teams in weekly operations reviews to identify feature gaps, assess upstream/downstream impacts, and implement automation solutions

Unisys

Lead System Engineer

March 2020 - June 2022 (2 years 4 months)

- Led end-to-end design and development of enterprise web applications aligned to customer requirements across global engagements
- Developed proof-of-concepts (POCs) evaluating emerging technologies and driving innovation within the software development lifecycle
- Conducted code reviews, unit testing, and system verification ensuring defect-free, high-performance application delivery
- Prepared high-level and detailed design documents ensuring full traceability and clarity for implementation teams
- Maintained Requirements Traceability Matrix (RTM) supporting quality assurance and audit readiness
- Acted as liaison between development and customer support teams, ensuring smooth issue resolution and operational alignment

Cognizant

Senior Associate

September 2014 - March 2020 (5 years 7 months)

Connecticut, United States

- Designed and developed integration layers between client legacy systems and Cognizant's TranZform portal using Open API specifications, enabling seamless data exchange across enterprise platforms
- Optimized database stored procedures and performed query performance tuning, achieving 25% improvement in data operation efficiency

- Authored high-level and detailed design documents aligned to business and architectural standards across multiple client engagements
- Maintained Requirements Traceability Matrix (RTM) ensuring full test coverage, validation, and audit readiness
- Led Agile sprints and coordinated onshore/offshore teams to align delivery goals and resolve cross-timezone dependencies
- Conducted code reviews and unit testing to ensure high-quality, maintainable deliverables across a 5+ year client engagement

Source Edge · Arctern · Zenith Software · Indigo · Indian-SEO Software Engineer

January 2009 - August 2012 (3 years 8 months)

Bengaluru

Built full-stack web applications across multiple early-career engagements in the US and India. Delivered backend HTTP handlers, SQL stored procedures, and database systems for mobile and web clients. Integrated Google Maps navigation, PayPal payment gateway (Authorize & Capture), and location-based search features. Configured Ektron CMS widgets, page builder functionality, and dynamic flex navigation menus. Performed rigorous unit and negative testing across all delivered systems. These roles established a strong foundation in end-to-end web development, database design, and quality engineering practices.

Education

Biju Patnaik University of Technology, Odisha

Bachelor of Technology (B.Tech.), Computer Software Engineering · (June 2004 - August 2008)